

1.Butterworth IIR Filters:

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB:

PROGRAM:

```
clc;
```

```
clear;
```

```
close all;
```

```
% EX NO: 1
```

```
% Butterworth IIR Low-Pass Filter Design and Analysis
```

```
% DATE:
```

```
Fs = 10000;    % Sampling Frequency (Hz)
```

```
Fp = 1000;    % Passband Frequency (Hz)
```

```
Fst = 1500;    % Stopband Frequency (Hz)
```

```
% Determine minimum filter order
```

```
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
```

```
% Design Butterworth Low-Pass Filter
```

```
[b, a] = butter(N, Wn, 'low');
```

```
% Frequency Response
```

```
figure;
```

```
freqz(b, a);
```

```
title('Frequency Response of Butterworth IIR Low-Pass Filter');
```

```
% Impulse Response
```

```
figure;
```

```
[h, n] = impz(b, a, 50);
```

```
stem(n, h, 'filled');
```

```
grid on;
```

```
title('Impulse Response');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
% Step Response
```

```
figure;
```

```
u = ones(1, 50);      % Unit step input
```

```
y = filter(b, a, u);
```

```
stem(0:49, y, 'filled');
```

```
grid on;
```

```
title('Step Response');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);
```

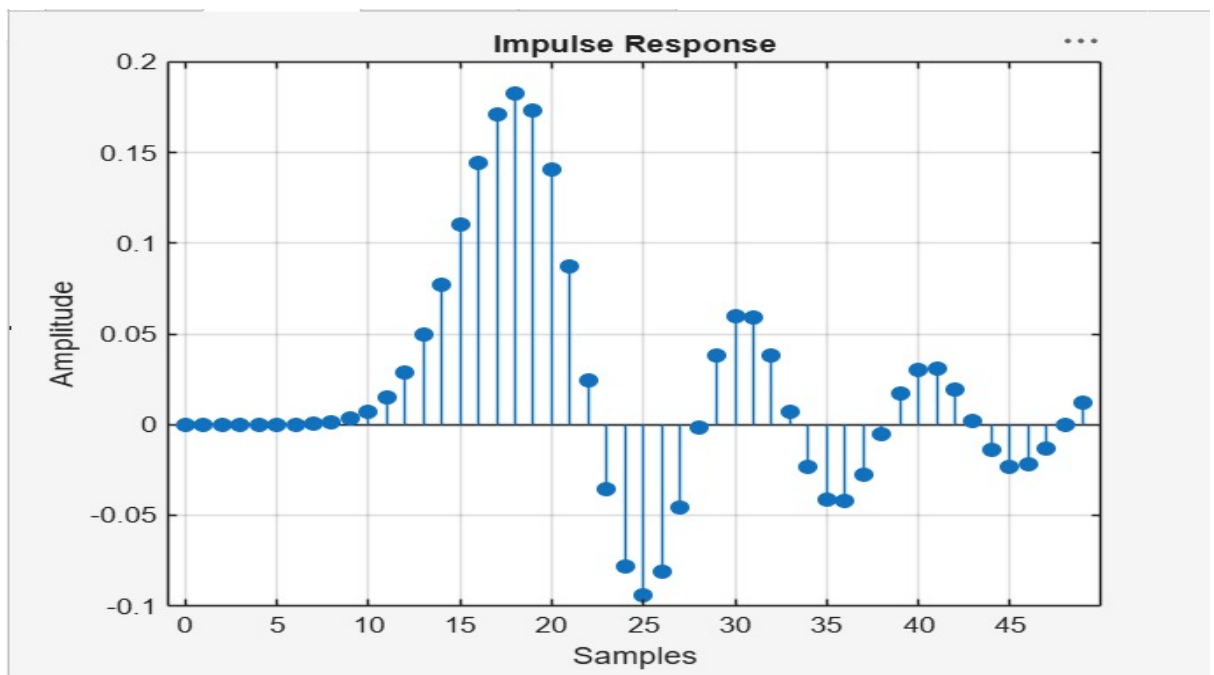
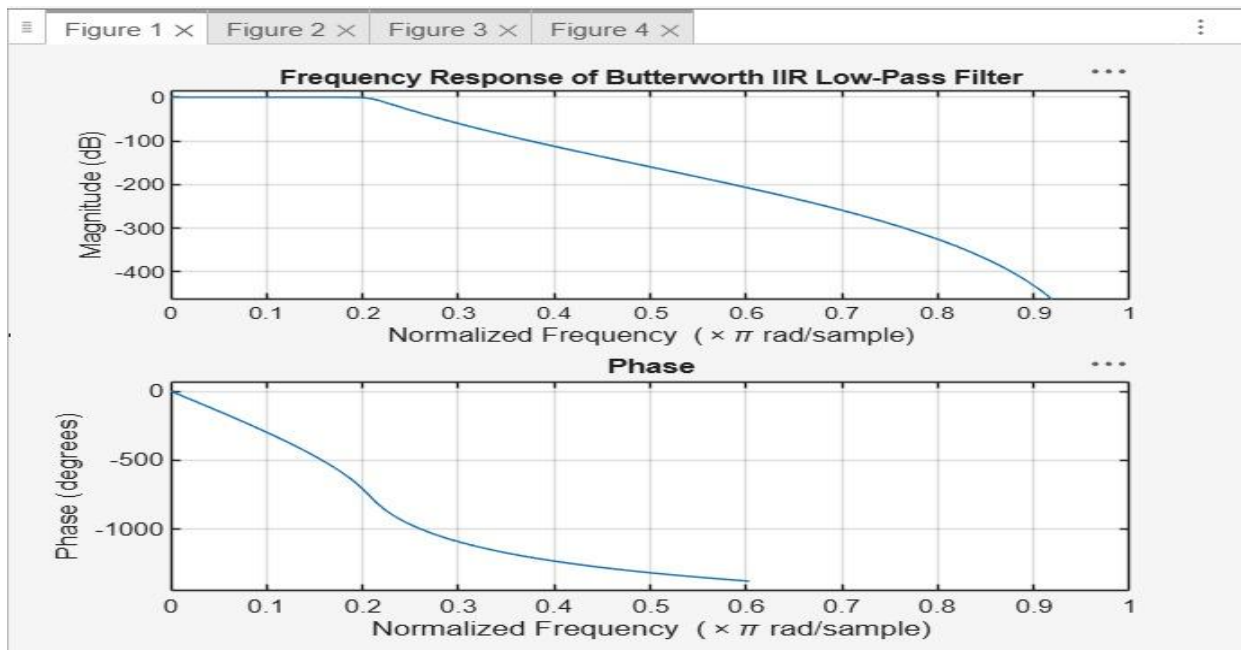
```
title('Pole-Zero Plot');
```

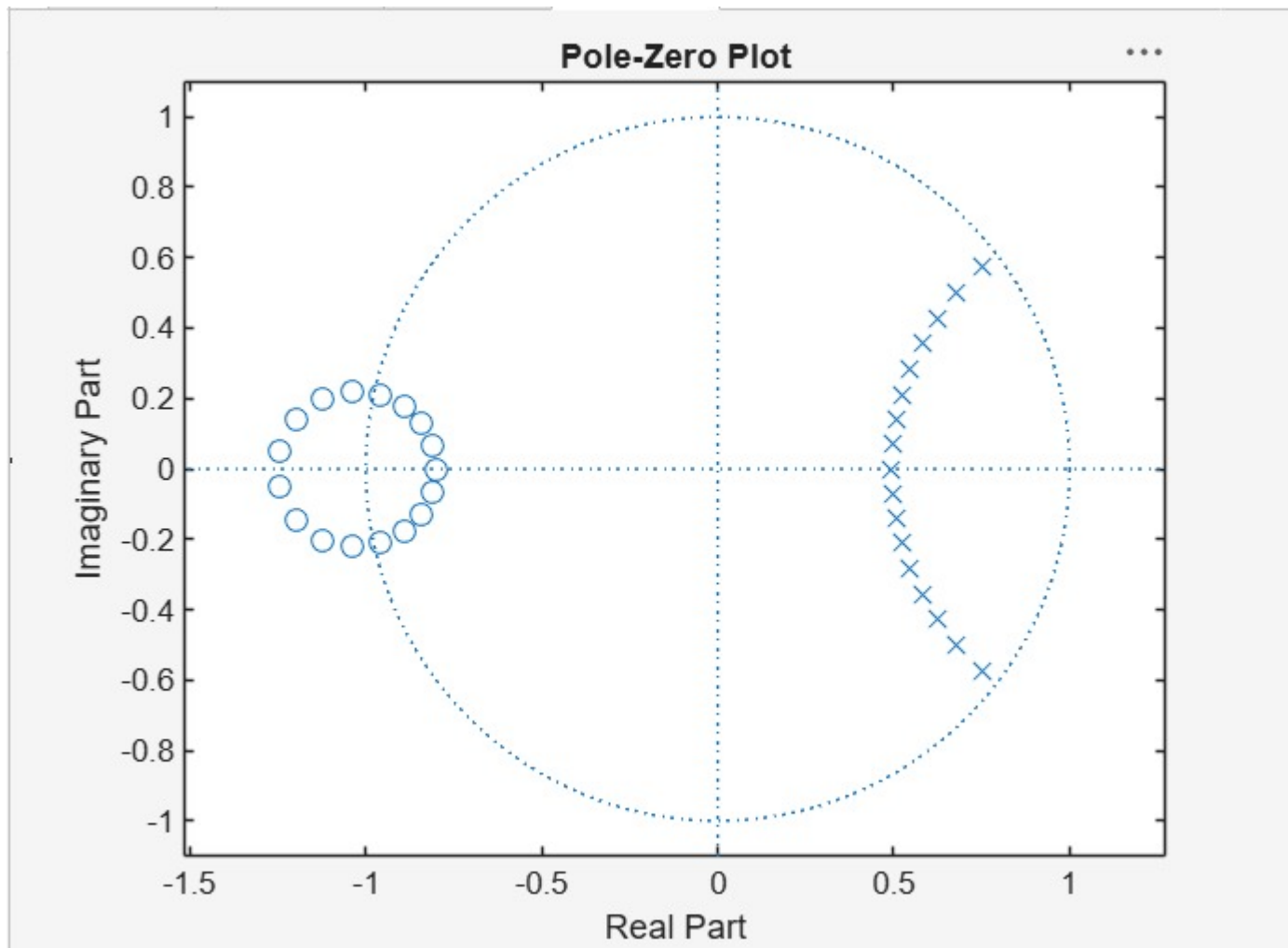
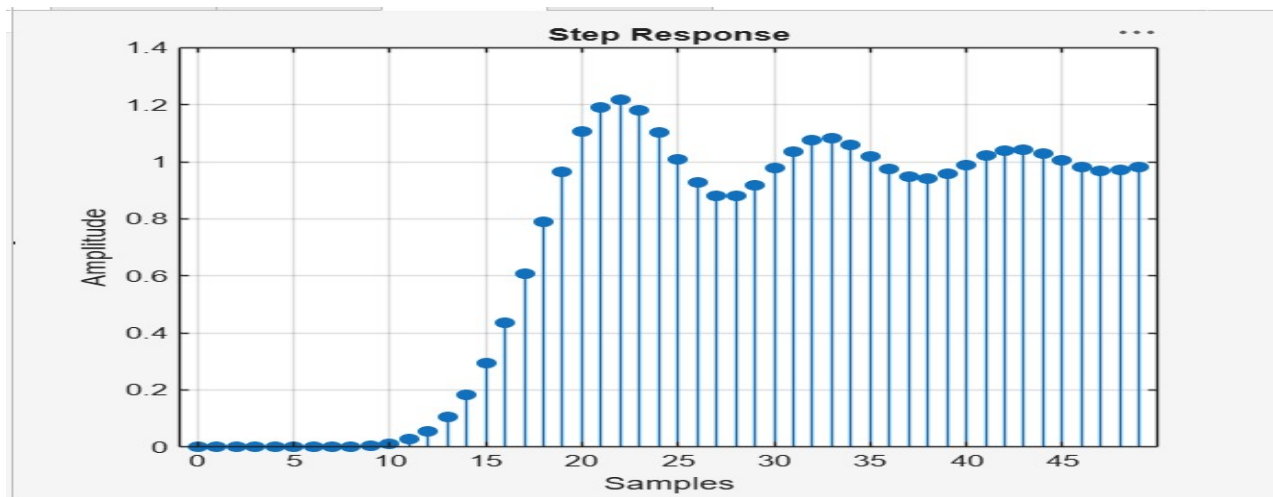
```
% Display Results
```

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

OUTPUT:





Minimum Filter Order (N) = 17
 Cutoff Frequency (Wn) = 0.2083
 >>

1.2. Design and Analysis of a Butterworth IIR High-Pass Filter:

Using MATLAB:

PROGRAM:

```
clc; clear
```

```
Fs=10000; Fp=3000; Fst=2000;
```

```
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60); [b,a]=butter(N,Wn,'high');
```

```
% Magnitude & Phase
```

```
freqz(b,a)
```

```
% Impulse Response
```

```
figure
```

```
[h,n]=impz(b,a,50);
```

```
stem(n,h), grid on
```

```
title('Impulse Response')
```

```
% Step Response
```

```
figure
```

```
stepz(b,a)
```

```
title('Step Response')
```

```
% Group Delay
```

```
figure
```

```
grpdelay(b,a)
```

```
title('Group Delay')
```

```
% Pole-Zero Plot
```

```
figure
```

```
zplane(b,a)
```

```
title('Pole-Zero Plot')
```

```
fprintf('Minimum Filter Order (N) = %d\n',N); fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

OUTPUT:

